Radionuclide Identification for Plastic Scintillation Detectors by Backward Cumulative Channel Sum

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ABSTRACT

Accurate identification of radionuclide using plastic scintillation detectors is crucial for screening applications, particularly in radiation portal monitors (RPMs). We present a backward cumulative channel sum (BCCS) method for analyzing gamma-ray spectra acquired with plastic scintillation detectors. The BCCS method transforms pulse-height spectra into cumulative distributions to reduce statistical fluctuations and enhance spectral contrast. This is especially effective under high background and low net count conditions where signals marginally exceed background levels. The method was validated with spectra for single and mixed radionuclides in screening scenarios, showing appropriate performance in isolating radionuclides. The BCCS provides interpretable results with light computational loads, making it suitable for embedded or mobile scanning systems. It can also be directly applied to existing RPMs without additional equipment.

Introduction

previous

Radiation portal monitors (RPMs) deployed at national borders play an important role in preventing the smuggling of nuclear and radioactive materials. These systems must detect weak radiation signals from high background radiation counts, requiring high detection efficiency. To enhance gamma-ray detection efficiency, two principal approaches are typically considered: employing high atomic number (high-Z) materials to increase the interaction probability, or increasing the detector volume to improve geometric efficiency. In practice, the latter approach is often implemented using plastic scintillators, which are widely utilized in radiation portal monitors (RPMs) due to their affordability and ease of manufacturing into large-scale detectors.[1,2]

Plastic scintillators have poor energy resolution and absence of full energy peaks in their spectra. Therefore, conventional spectroscopic analysis cannot be applied to the plastic. Furthermore, most of counts are concentrated in the low-energy channels[1], making it difficult to distinguish radionuclides signals from background when the signal intensity is slightly higher than background.

Recent studies have addressed this challenge using spectral data processing methods, including the energy window methods[3,4], which defines regions-of-interest for each radionuclide, the energy-weighted method[5], which emphasizes spectral shape near Compton Edges. There are several template matching techniques, e.g., spectral angular mapping[6] and inverse calibration matrix[7], which compare measured data with pre-defined radionuclide dataset. In parallel, Kim et al. demonstrated the feasibility of artificial neural network-based radionuclide identification[8], and another group applied energy-weighted preprocessing combined with convolutional neural networks to improve radionuclide identification performance in RPM applications[9].

Building on these advances, our previous work introduced artificial intelligence (AI)-driven methods for plastic scintillation detectors. These include a machine learning approach[10], a multitask deep learning model for pseudo-gamma spectroscopy[11], and a deep learning model with its explanation[12]. These approaches demonstrated improved nuclide identification, even in the absence of full-energy peaks.

In practical settings, additional challenges arise. Cargo containers often induce baseline suppression—also known as the shadow effect—where both background and source counts are significantly reduced. Observational studies report count losses exceeding 30%, depending on vehicle geometry and panel position[13]. Such suppression degrades gross-count or peak-based methods and reduces identification confidence, especially in low-count scenarios.

This paper presents a backward cumulative channel sum (BCCS) to address these limitations. BCCS transforms raw plastic scintillator spectra into backward cumulative form, emphasizing global spectral structure while reducing high-frequency uncertainties. Unlike conventional background subtraction or AI-based methods that require extensive training or calibration, BCCS is computationally lightweight, interpretable, and well suited for real-time embedded systems. Its performance was evaluated through experiments with single and multiple radionuclides, and it was verified that the BCCS enables accurate and stable nuclide identification—even under low-count and baseline-suppressed conditions—without relying on peak-based features or machine learning. This work provides a practical, scalable, and calibration-free solution for radionuclide identification in radiation portal monitoring systems using plastic scintillators.

current

Radiation portal monitors (RPMs) are widely deployed at border crossings, transportation facilities, and industrial sites to detect illicit transportation of radioactive materials[1–3]. Most RPM systems utilize large-area plastic scintillation detectors because of their low cost, mechanical robustness, fast response, and high detection efficiency[4–6]. However, plastic scintillators inherently exhibit poor energy resolution, and the resulting spectra are dominated by broad Compton continua rather than well-defined photopeaks. This limitation makes radionuclide identification particularly challenging under field conditions, where the net count rate is often low and background radiation can vary significantly over time.

To improve radionuclide identification using low-resolution detectors, a variety of signal-processing and spectral-analysis techniques have been proposed. Traditional approaches include smoothing or filtering methods [7], feature extraction techniques such as principal component analysis (PCA) [8], and pattern-recognition algorithms based on machine learning [9–11]. Although these methods have demonstrated performance improvements in controlled settings, many require substantial training datasets, rely on stable environmental conditions, or are sensitive to background fluctuations. Such constraints limit their practicality for real-time operation in RPM systems deployed in diverse and dynamic environments.

In typical RPM operation, radioactive sources must often be detected at large source-to-detector distances or through shielding materials, conditions that lead to extremely low net-count measurements. Under these circumstances, conventional approaches tend to lose sensitivity because spectral features become obscured by statistical fluctuations. Therefore, there is a strong need for preprocessing techniques that enhance spectral separability without relying on high-resolution detectors or computationally intensive algorithms.

In this work, we propose the Backward Cumulative Channel Sum (BCCS) as a simple yet robust spectral transformation method tailored for plastic scintillator–based RPM applications. The BCCS transformation suppresses statistical noise in the high-channel region while preserving essential spectral structure, thereby improving the distinguishability of radionuclides under low-count conditions. To further isolate source-specific features, we apply two background-compensation techniques: direct subtraction and background normalization, which respectively provide absolute and relative representations of source contributions. The processed spectra remain compatible with linear and nonlinear spectral-unfolding techniques, enabling quantitative estimation of radionuclide contributions in mixed-source measurements. Using experimental data from Am-241, Co-60, and Cs-137 across various source-to-detector distances, we demonstrate that the proposed framework supports both qualitative source identification and quantitative activity estimation using only low-resolution plastic scintillation detectors.

Methods

Spectral Preprocessing: Backward Cumulative Channel Sum

Most radiation portal monitors employ plastic scintillators, which provide high detection efficiency due to their large effective volume but suffer from poor spectroscopic performance. As a result, additional processing techniques are required to identify radionuclides, particularly when source intensity is lower than background level, i.e., under low net-count conditions. In such scenarios, signal-processing methods that enhance spectral features and compensate for detector limitations become essential. To address these challenges, we propose the Backward Cumulative Channel Sum (BCCS) method, derived as follows.

Given a raw pulse-height spectrum $\mathbf{C} = [C_1, C_2, \ldots, C_N]$, where C_i denotes the count in channel i, the corresponnding BCCS spectrum $\mathbf{B} = [B_1, B_2, \ldots, B_N]$ is defined as follows:

$$B_i = \sum_{j=i}^{N} C_i. \tag{1}$$

In contrast to the conventional cumulative spectrum, which sums counts from the first channel up to channel i, the BCCS accumulates counts from channel i to the final channel. As a result, the BCCS spectrum is monotonically decreasing distributions in which each bin represents the total counts from that bin to the end of the spectrum. The BCCS transformation effectively suppresses statistical fluctuations in higher-energy channels while preserving the overall spectral shapes, making it particularly useful low net-count measurements in RPM applications.

Background Compensation Techniques

Under weak source or short acquisition conditions, measured spectra are dominated by background radiation and often exhibit poor counting statistics. To improve the signal-to-noise ratio and enable reliable radionuclide identification, it is necessary to compensate for background contributions in the measured spectra. In this work, we employ two complementary background-compensation strategies: (a) direct subtraction and (b) background normalization, each offering distinct advantages in terms of interpretability and robustness.

(a) Direct Subtraction Method

Direct subtraction obtains the net spectrum by subtracting the background BCCS spectrum from that of the source. Let $\mathbf{B}$ and $\mathbf{B}^{\text{bkg}}$ denote the BCCS spectra of source and background, respectively. The net spectrum $\mathbf{S} = [S_1, \ldots, S_N]$ is then defined as

$$S_i = B_i - B_i^{\text{bkg}}. \tag{2}$$

This operation provides an interpretable estimate of the source contribution while preserving the overall count scale. However, due to the cumulative nature of BCCS, small negative values may appear as a result of statistical fluctuations, particularly when counts are concentrated in the low-channel region or when the overall count rate is extremely low. In the latter case, radionuclide analysis becomes practically meaningless, whereas the former situation will be revisited and discussed later when analyzing experimental spectra.

(b) Ratio Method : Normalization to background

Ratio-based normalization highlights relative spectral deviations by dividing the source BCCS spectrum by the background spectrum. Therefore, the normalized spectrum $\mathbf{R} = [R_1, \ldots, R_N]$ is defineds as

$$R_i = \left(\frac{B_i}{B_i^{\text{bkg}}}\right) - 1 = \frac{S_i}{B_i^{\text{bkg}}}. \tag{3}$$

This produces a dimensionless representation that emphasizes local contrasts while suppressing global count variability. As a result, this method is particularly effective when source signals only slightly exceed above background levels.

Complementary Roles of the Two Methods

Direct subtraction provides an absolute estimate of the source contribution, whereas the ratio method highlights relative deviations from background. Combining these complementary perspectives ensures flexibility and robustness across diverse measurement conditions, including low-count scenarios and mixed-source environments.

Radionuclide Activity Estimation via Linear Spectral Unfolding

Let $\mathbf{C}^X$ and $\mathbf{C}^{X,\text{bkg}}$ denote the reference spectrum fof radionuclide X and its corresponding background, respectively. Given a measure spectrum $\mathbf{C}^M$ and its background $\mathbf{C}^{M,\text{bkg}}$, our goal is to determine which radionuclides are present in what proportions. For quantitative analysis, the measured spectrum can be modeld as a linear combination of baccground-subtracted reference spectra :

$$\mathbf{C}^M \approx \sum_X \alpha_X (\mathbf{C}^X - \mathbf{C}^{X,\text{bkg}}) + \mathbf{C}^{M,\text{bkg}} \tag{4}$$

where α_X represents the contribution of radionuclide X.

Because the BCCS transformation preserves linearity, the same relation holds after applying direct subtraction. The BCCS-transformed and background-subtracted measure spectrum $\mathbf{S^M}$ satisfies

$$S_i^M = \sum_X \alpha_X S_i^X. \tag{5}$$

where S_i^X is the background-subtracted BCCS spectrum of radionuclides X. Similary, for the ratio-normalized spectra, we obtain

$$R_i^M = \sum_X \alpha_X \left(\frac{S_i^X}{B_i^{M,\text{bkg}}}\right).$$

If the background distributions of the measured spectrum and reference spectra are compareble, i.e., $B_i^{M,\text{bkg}} \approx B_i^{X,\text{bkg}}$, then

$$R_i^M = \sum_X \alpha_X \left(\frac{S_i^X}{B_i^{X,\text{bkg}}}\right) = \sum_X \alpha_X R_i^X. \tag{6}$$

In both cases-direct subtraction (5) and ratio normalizaiton (6)-the coefficients α_X can be estimated by solving the non-negative least squares (NNLS) problem :

$$\underset{\alpha_X \geq 0}{\arg\min} \left\| \mathbf{I}^{\mathrm{M}} - \sum_{X} \alpha_X \mathbf{I}^{X} \right\|^2, \tag{7}$$

where $\mathbf{I}^{\mathbf{M}}$ and $\mathbf{I}^{\mathbf{X}}$ denote the measured and reference spectra after applying the chosen background compensation method.

This procedures provide physically meaningful, non-negative estimates of radionuclide contributions, even in the presence of spectral overlap, low counts, or poor energy resolutions. By appying NNLS to both direct-subtraction spectra dn ratio-normalized spectra, reliable decomposition of mixed-source measurements can be achieved.

Results

Spectral Transformation of Reference Radionuclides Spectra

Figure 1 (a) presents the raw spectra acquired for background radiation and three reference radionuclides—Am-241, Co-60, and Cs-137—using a large-volume plastic scintillation detector (30 cm ×× 180 cm × 5 cm). Each of Co-60 and Cs-137 was measured at two different source-to-detector distances to illustrate the expected attenuation effects. As typical for plastic scintillators, all spectra exhibit broad Compton continua with poor energy resolution, resulting in most counts being concentrated in the low-channel region (channels < 100). Co-60 and Cs-137 show distinct deviations from the background spectrum across a broader channel range, whereas Am-241 exhibits only a slight difference confined to the very low-channel region (inset of Figure 1 (a)). As anticipated, these differences diminish as the source-to-detector distance increases. For subsequent analysis, the spectra measured at the shorter distances were selected as reference data. Application of the BCCS method transforms the raw spectra into smooth, monotonically decreasing cumulative distributions, as shown in Figure 1 (b). This transformation effectively suppresses statistical fluctuations in the high-channel region while preserving the overall spectral structure, providing a more stable representation for subsequent processing.

Background compensation

Two background compensation methods—direct subtraction and ratio-based normalization—were applied to the BCCS-transformed spectra to improve separation between background and source contributions. The direct subtraction method effectively isolated the net source signal by removing the background BCCS spectrum, producing interpretable cumulative profiles that preserved radionuclide-specific spectral characteristics (Figure 1 (c)). This behavior remained consistent in mixed-source measurements, where the resulting spectra maintained mixture-dependent features that reflect the relative contributions of individual radionuclides (Figure 2 (a)).

In the case of Am-241, the spectrum is concentrated almost entirely in the very low-channel region. As a result, statistical fluctuations can generate small negative values in the background-subtracted BCCS spectrum. These negative values are less than one-tenth of the maximum amplitude of the Am-241 signal and therefore contribute negligibly to the overall spectral shape. Since they arise purely from statistical noise rather than any physical process, they are set to zero in the subsequent fitting procedure to prevent spurious effects in the quantitative analysis.

The ratio-based normalization method (Figure 1 (d)) emphasized proportional deviations from the background spectrum, revealing characteristic source-induced patterns even under low-count conditions. By suppressing global count-level variations, this method enhanced the visibility of subtle spectral differences and improved pattern recognition when source signals only slightly exceeded background levels.

Together, these two approaches provide complementary perspectives: direct subtraction captures absolute source contributions, while ratio normalization highlights relative contrasts. Their combined use supports robust spectral interpretation across a wide range of measurement scenarios, including low-count and mixed-source environments.

Spectral unfolding and Radioisotope identification

Figure 2(a) presents the mixed-source spectra obtained using the background-subtracted BCCS method, while Figure 2(b) shows the corresponding results produced by background normalization. A comparison of the two approaches indicates that the background-normalized spectra provide a more intuitive visual representation of source presence, particularly when the overall radiation intensity is relatively high or when the constituent radionuclides contain Compton features in different channel regions. In such cases, the ratio-normalized spectra accentuate these differences more distinctly, enabling mixed-source compositions to be recognized more readily. This enhanced visual contrast also facilitates subsequent quantitative decomposition through linear spectral unfolding.

Applying nonlinear least-squares fitting to Eq. 7 enables estimation of the contribution of each radionuclide for both background-compensation methods. Due to the low net counts and the absence of distinctive spectral features in the high-channel region, only the low-channel range ($0 \leq i \leq 100$) is used for the fitting. Figures 3 and 4 present the fitting results for the background-subtracted and background-normalized spectra, respectively, each illustrating the estimated contributions of the reference radionuclides.

Figure 5 summarizes the estimated contributions as a function of source-to-detector distance. The horizontal dashed lines indicate the mean contributions obtained from samples that do not contain the corresponding radionuclide. The monotonic decrease in contribution with increasing distance aligns with general expectations based on geometric attenuation. As observed for Co-60 and Cs-137, when the activity of a radionuclide becomes sufficiently low, its contribution can no longer be reliably determined using this method. These results demonstrate that the proposed data-processing techniques enable effective radionuclide identification and quantitative activity estimation for plastic scintillation detectors.

Discussion

This study investigated spectral data processing methods to improve radionuclide identification performance using plastic scintillation detectors for RPMs or food screening systems. The proposed methods—BCCS, direct subtraction, and ratio-based normalization—were validated through experiments involving both single and mixed radionuclides.

The BCCS method demonstrated its effectiveness in reducing high-frequency statistical uncertainties while enhancing overall spectral contrast, enabling more reliable analysis of low-resolution spectral data. Direct subtraction provided interpretable net spectra by isolating source contributions from background, while ratio-based normalization emphasized relative spectral deviations, improving pattern recognition under low-count or variable background conditions.

Both methods retained distinct spectral features in mixed-source scenarios, supporting quantitative analysis through linear spectral unfolding. The NNLS method enabled the quatitative estimation of individual radionuclide contributions without relying on energy peak resolution, one of the limitations of plastic scintillation detectors.

This capability extends to broader applications such as border security, cargo inspection, scrap metal recycling, and environmental radiation monitoring, where differentiating between anthropogenic and naturally occurring radiation is critical.

In summary, the complementary strengths of direct subtraction and ratio-based normalization suggest that a hybrid approach may offer optimal performance across diverse operational scenarios. Future work will focus on integrating these methods into field-deployable RPM systems, validating performance in dynamic environments, and extending the framework to incorporate real-time background modeling and adaptive detection algorithms.

Conclusion

This study presented a robust and efficient methodology for radionuclide identification for plastic scintillation detectors in radiation portal monitoring and food safety applications. The proposed methods—Backward Cumulative Spectral Normalization, direct subtraction, and ratio-based normalization—were systematically validated by simulations and experiments. Key findings of this study include:

• Enhanced Spectral Robustness: BCCS effectively suppressed statistical uncertainties while preserving global spectral features.

• Reliable Background Compensation: Direct subtraction and ratio-based normalization enabled accurate separation of source signals from background radiation.

• Quantitative Analysis Capability: Linear spectral unfolding with NNLS provided physically meaningful estimates of mixed radionuclide contributions.

These results confirm that the proposed methods offer a scalable and interpretable radionuclide identification solution for radiation detectors with poor energy resolution. Future research will aim to validate these techniques in various operational scenarios and enhance their integration into next-generation radiation monitors.

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Author contributions statement

M. Moon conceived the study and performed the experiments. J. So developed the analysis software and performed signal processing. B. Jeon synthesized the test datasets and supported evaluation. All authors contributed to writing and reviewing the manuscript.

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Additional information

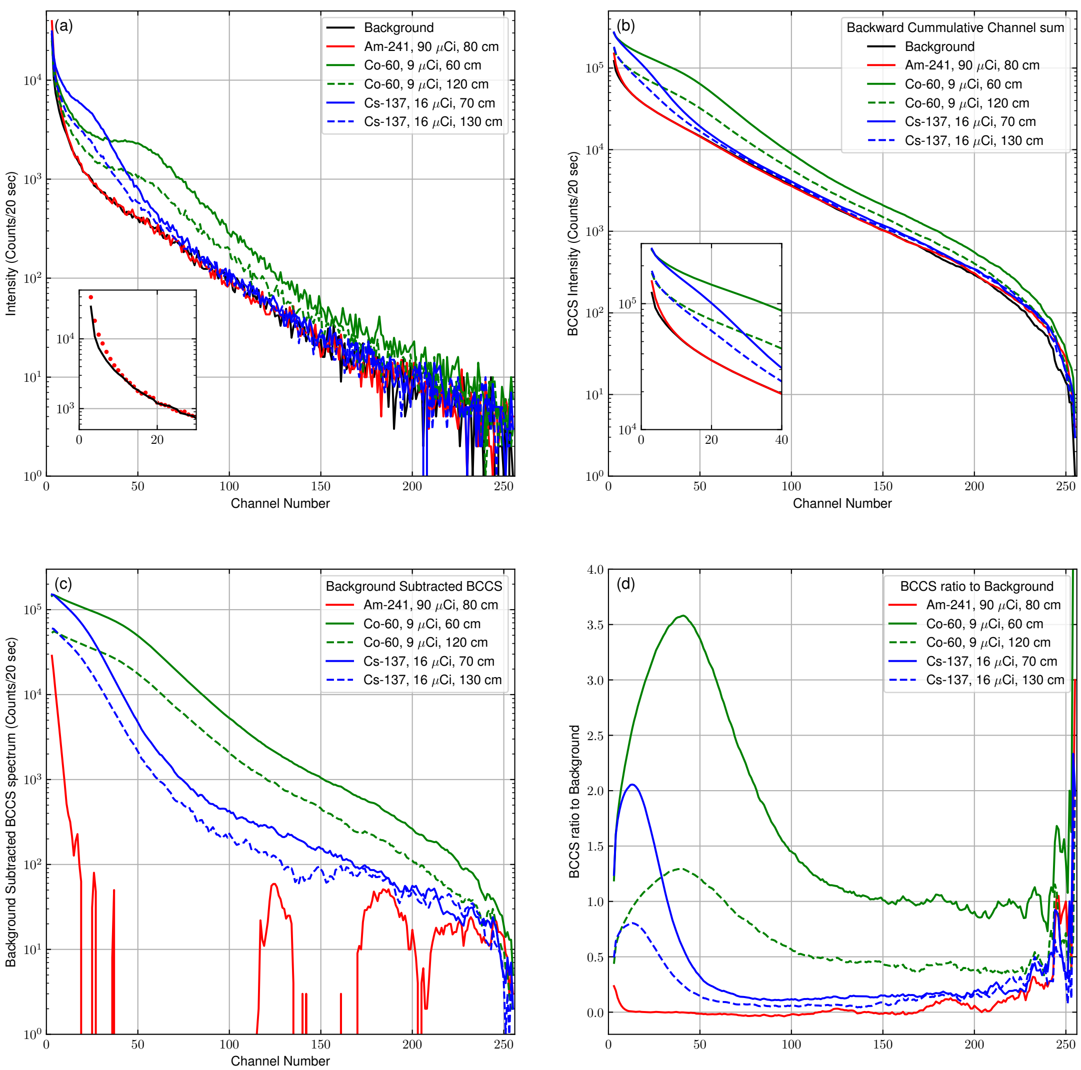

Figure 1. (a) Raw gamma-ray spectra acquired from background (black) and reference radionuclides—Am-241 (red), Co-60 (green), and Cs-137 (blue). Source-to-detector distances are indicated in the legend; Co-60 and Cs-137 were measured at two different distances to illustrate distance-dependent attenuation effects. The shoter and longer source-to-detector distance are represented as solid and dashed lines. The inset highlights the marginal difference between the background and Am-241 spectra, visible only in the low-channel region. (b) BCCS-transformed spectra of background and reference radionuclides. The inset shows difference of BCCS spectrum at low channel region. (c) Background-subtracted BCCS spectra of single radionuclides obtained by subtracting the background BCCS spectrum from each source spectrum. (d) Ratio-normalized spectra of single radionuclide obtained by computing the channel-wise ratio of the source-plus-background spectrum to the background spectrum.

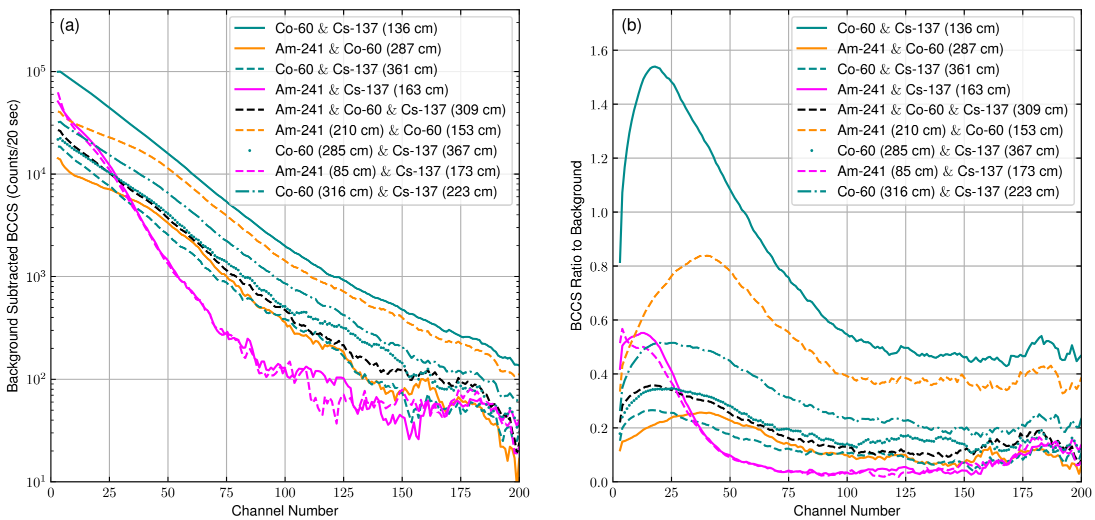

Figure 2. Background-subtracted BCCS spectra(a) and Background-normalized BCCS ratio(b) of nine mixed radionuclide configurations. The mix composition and source-to-detector distances are indicated in the legend. The first five configurations share the same source-to-detector distance, whereas the remaining four were measured at different distances. The cumulative profiles retain mixture-specific spectral features, enabling quantitative analysis through linear spectral unfolding.

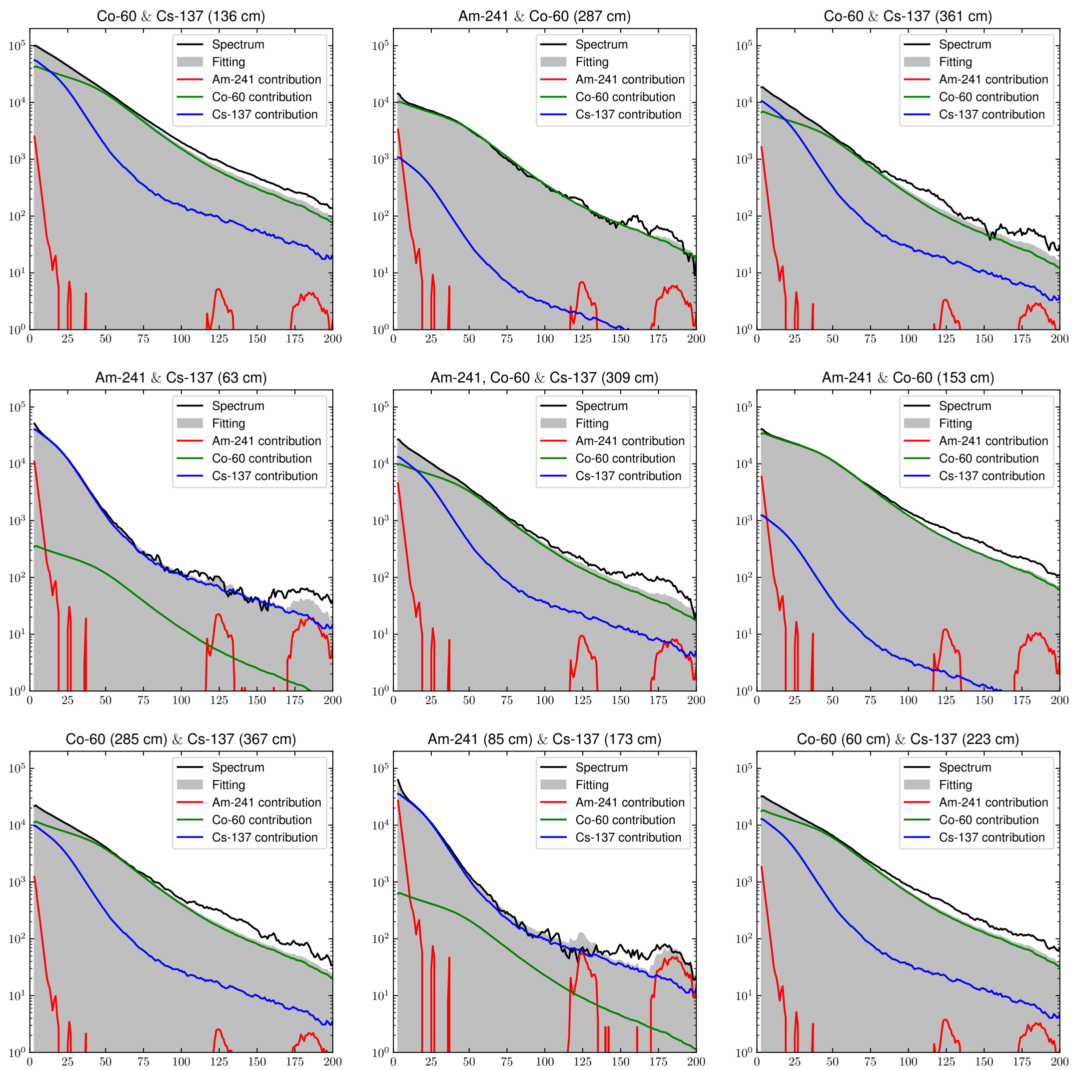

Figure 3. Fitting results for bacground subtracted method. The gray filled plot is the sum of estimated contribution of three radioisotpoes.

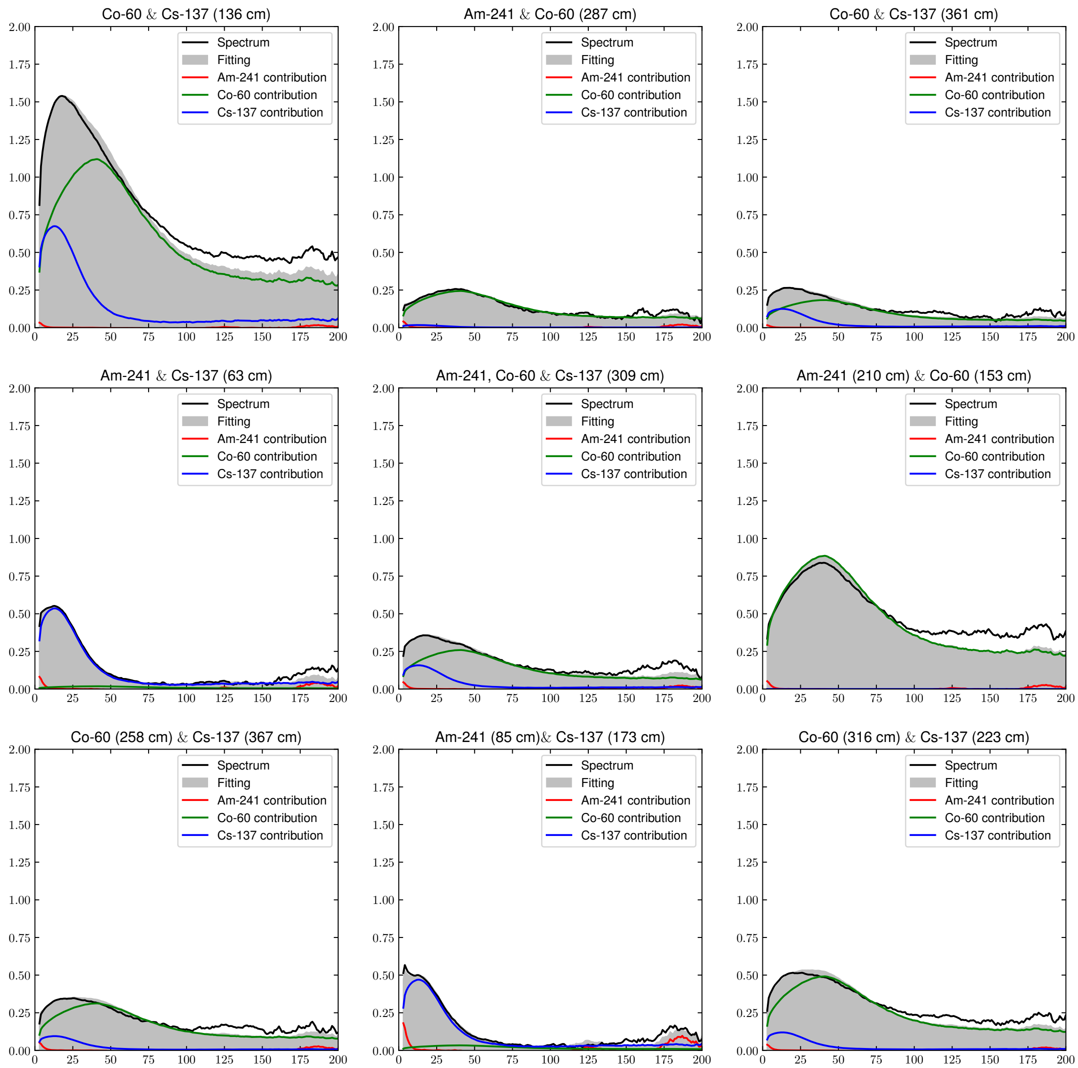

Figure 4. Fitting results for bacground normalization method. The gray filled plot is the sum of estimated contribution of three radioisotpoes.

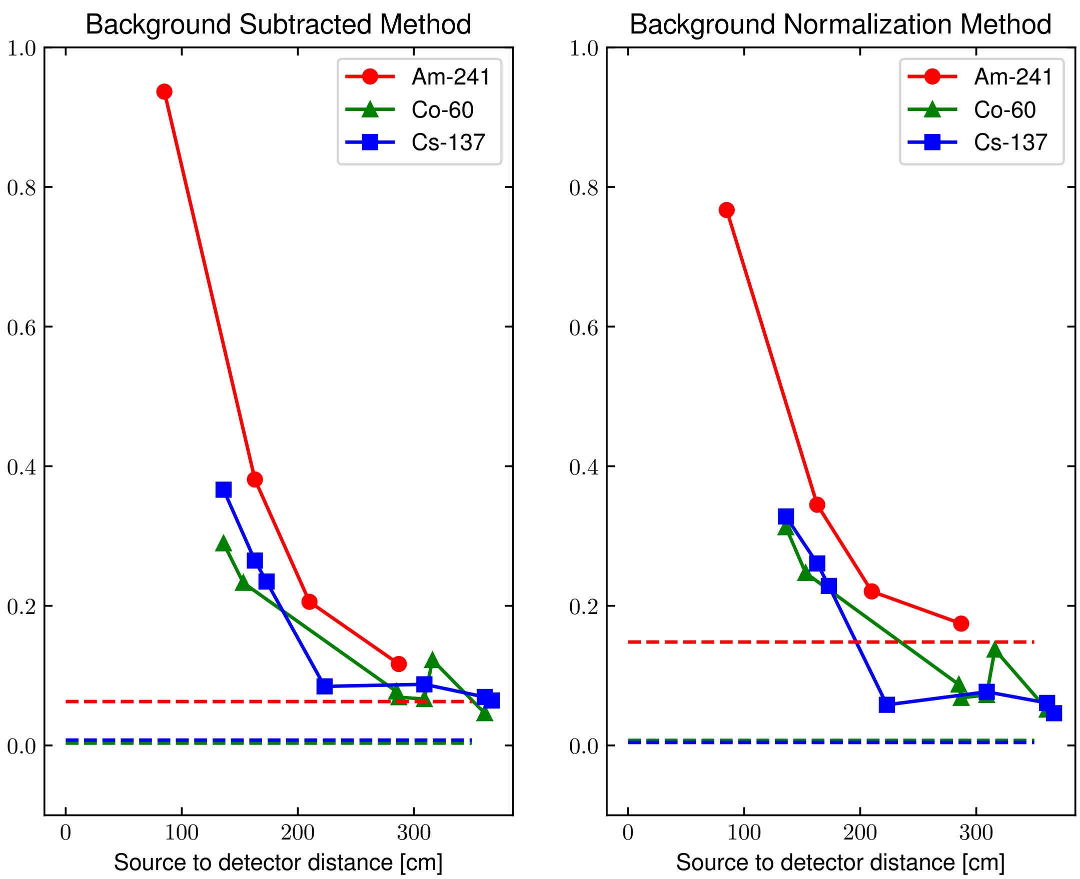

Figure 5. Estimated contribution of radioisotopopes for the sample to detector distance. The horizontal dashed lines indicate the estimated contributions obtained from samples that do not contain the corresponding radionuclide.